

Problem

Let $\mathbb{M}_{22}$ be the space of all 2×2 matrices and let $\mathbb{P}_2$ be the space of all polynomials of at most degree 2. Define $S : \mathbb{P}_2 \rightarrow \mathbb{M}_{22}$ as the transformation given by

$$S(a_2x^2 + a_1x + a_0) = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix},$$

for all $a_2x^2 + a_1x + a_0 \in \mathbb{P}_2$.

- 1 Describe the null space of S . Is S one-to-one?
- 2 Express the range of S as a span of a linearly independent set of 2×2 matrices. Is S onto?
- 3 Is S an isomorphism?
- 4 Extend the basis of the range of S such that the extended basis, $\mathcal{B}_1$, spans $\mathbb{M}_{22}$.
- 5 Consider the ordered basis $\mathcal{B} = \{1, x, x^2\}$ and the ordered basis $\mathcal{B}_1$ for $\mathbb{M}_{22}$ and find the matrix of the transformation S .
- 6 Consider the ordered basis $\mathcal{B} = \{1, x, x^2\}$ and the ordered basis $\mathcal{B}_2 = \left\{ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ for $\mathbb{M}_{22}$ and find the matrix of the transformation S .

$$1 \quad \text{Nul } S = \{P(x) = a_2x^2 + a_1x + a_0 : S(P(x)) = O_{2 \times 2}\}$$

$$S(P(x)) = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{cases} a_1 + a_2 = 0 \\ a_2 + a_0 = 0 \\ a_1 - a_0 = 0 \\ a_1 + a_0 = 0 \end{cases} \rightarrow a_2 = a_1 = a_0 = 0$$

which is the trivial solution.

Hence, null space only contains the zero vector of $\mathbb{P}_2$ which is known as the zero polynomial $p_0(x) = (0)x^2 + 0(x) + 0(1) = 0$ and S is one-to-one.

$$\boxed{\text{Nul } S = \{p_0(x)\}}$$

2 The output of S is $\begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix} = a_0 \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}}_{M_0} + a_1 \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_{M_1} + a_2 \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}}_{M_2}.$

we can check that $\{M_0, M_1, M_2\}$ is a linearly independent set and hence this set is a basis for range of S , since range of S is $\text{span}\{M_0, M_1, M_2\}$. This means $\dim(\text{range of } S) = 3$, whereas $\dim(M_{22}) = 4$. Therefore, S is NOT onto.

3 Is S an isomorphism? No. S is not onto.

4 Extend the basis of the range of S such that the extended basis, $\mathcal{B}_1$, spans $\mathbb{M}_{22}$.

$$\{M_0, M_1, M_2\} = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\}$$

and we want to add a fourth matrix, M_3 , such that M_3 cannot be written as $M_3 = a_0 M_0 + a_1 M_1 + a_2 M_2$. In other words, the following must not have a solution:

$$M_3 = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix}$$

As a system of linear eqs. we have:

$$\begin{array}{c} a_2 \quad a_1 \quad a_0 \\ \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 1 & 0 & 1 & b_2 \\ 0 & 1 & -1 & b_3 \\ 0 & 1 & 1 & b_4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & -1 & 1 & b_2 - b_1 \\ 0 & 1 & -1 & b_3 \\ 0 & 1 & 1 & b_4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & b_4 \\ 0 & 1 & -1 & b_3 \\ 0 & -1 & 1 & b_2 - b_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & b_4 \\ 0 & 1 & -1 & b_3 \\ 0 & 0 & 0 & b_3 + b_2 - b_1 \end{array} \right] \end{array}$$

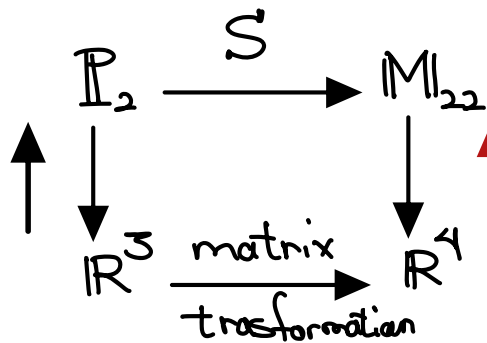
If $b_3 + b_2 - b_1 \neq 0$, Then the system is inconsistent. For example, choose

$b_1 = 1, b_2 = b_3 = b_4 = 0$, which gives $M_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

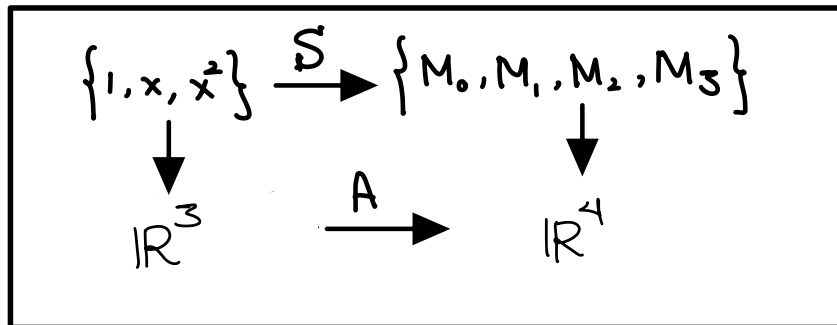
Therefore, the set

$B_1 = \{M_0, M_1, M_2, M_3\} = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$
is a basis for M_{22} .

- 5 Consider the ordered basis $B = \{1, x, x^2\}$ and the ordered basis B_1 for M_{22} and find the matrix of the transformation S .



After finding the output of the matrix trans, we can use the isomorphism between M_{22} and R^4 and write the output as a 2×2 matrix.



$$S(a_2x^2 + a_1x + a_0) = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix}$$

$$a_0 S(1) + a_1 S(x) + a_2 S(x^2) = a_0 M_0 + a_1 M_1 + a_2 M_2 + 0 M_3$$

$$a_0 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + a_1 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ 0 \end{bmatrix}$$

$$\boxed{S(1) = M_0} \quad \boxed{S(x) = M_1} \quad \boxed{S(x^2) = M_2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{4 \times 3} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ 0 \end{bmatrix}_{4 \times 1}$$

Hence, the matrix of the transformation S with $\beta = \{1, x, x^2\}$ and $\beta_1 = \{M_0, M_1, M_2, M_3\}$ as ordered bases for $\mathbb{P}_2$ and M_{22} is

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{4 \times 3}.$$

As can be seen above, to find A_1 , we had to find $S(1)$, $S(x)$, $S(x^2)$ in terms of the basis elements M_0 , M_1 , M_2 , and M_3 . In other words, we needed to figure out how the basis elements of domain basis transform with respect to the basis β_1 of the codomain.

Here is how the elements of $\mathcal{B}$ are transformed:

$$S(1) = M_0 \quad S(x) = M_1 \quad S(x^2) = M_2$$

which means the matrix A is:

$$A = \begin{matrix} & \begin{matrix} 1 & x & x^2 \end{matrix} \\ \begin{matrix} M_0 \\ M_1 \\ M_2 \\ M_3 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

For example, $S(x) = M_1$ can be written as:

$$\begin{matrix} M_0 \\ M_1 \\ M_2 \\ M_3 \end{matrix} \begin{bmatrix} 1 & x & x^2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ x \\ x^2 \end{matrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} M_0 \\ M_1 \\ M_2 \\ M_3 \end{matrix}$$

- 6 Consider the ordered basis $\mathcal{B} = \{1, x, x^2\}$ and the ordered basis $\mathcal{B}_2 = \left\{ \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{M_{12}}, \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}}_{M_{21}}, \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{M_{22}}, \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{M_{11}} \right\}$ for $\mathbb{M}_{22}$ and find the matrix of the transformation S .

$$\mathcal{B}_1 = \{M_0, M_1, M_2, M_3\} = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

$$\begin{aligned} S(1) &= M_0 = M_{12} - M_{21} + M_{22} \\ S(x) &= M_1 = M_{21} + M_{22} + M_{11} \\ S(x^2) &= M_2 = M_{12} + M_{11} \end{aligned}$$

$$A_2 = \begin{matrix} & \begin{matrix} 1 & x & x^2 \end{matrix} \\ \begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

For example, $S(x) = M_{21} + M_{22} + M_{11}$

$$\begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix} \begin{bmatrix} 1 & x & x^2 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ x \\ x^2 \end{matrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix}$$

Alternatively, we can find a matrix that changes the basis β_1 to β_2 :

$$\begin{aligned} M_0 &= M_{12} - M_{21} + M_{22} \\ M_1 &= M_{21} + M_{22} + M_{11} \\ M_2 &= M_{12} + M_{11} \\ M_3 &= M_{11} \end{aligned}$$

$$P_{\beta_2 \leftarrow \beta_1} = \begin{matrix} M_0 & M_1 & M_2 & M_3 \\ \begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix} \begin{matrix} M_0 & M_1 & M_2 & M_3 \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix} \begin{matrix} M_0 \\ M_1 \\ M_2 \\ M_3 \end{matrix} \begin{bmatrix} 1 & x & x^2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{matrix} M_{12} \\ M_{21} \\ M_{22} \\ M_{11} \end{matrix} \begin{bmatrix} 1 & x & x^2 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$P_{\beta_2 \leftarrow \beta_1} \underbrace{A_1}_{S_{\beta, \beta_1}} = \underbrace{A_2}_{S_{\beta, \beta_2}}$$

$$\begin{aligned} A_1 [\vec{u}]_{\beta} &= [\vec{v}]_{\beta_1} \\ P_{\beta_2 \leftarrow \beta_1} A_1 [\vec{u}]_{\beta} &= P_{\beta_2 \leftarrow \beta_1} [\vec{v}]_{\beta_1} \\ \underbrace{P_{\beta_2 \leftarrow \beta_1} A_1}_{A_2} [\vec{u}]_{\beta} &= [\vec{v}]_{\beta_2} \end{aligned}$$

← Reminder